

YILDIZ TECHNICAL UNIVERSITY

Faculty of Education

Computer Networks & Operating Systems Joint Project

LearNexus

A mobile learning application for middle school students

Team Number	2
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Computer Networks Topic	Cybersecurity / Siber Güvenlik
Operating Systems Topic	File-System Interface / Dosya Sistemi Arayüzü
Instructor	Dr. Kiran
GitHub Repository Link	https://github.com/cemgedikoglu/LearNexus
Promotional Video Link	https://drive.google.com/file/d/1mEOLP_ZhbSZsaX5AvIFu4BuZIVReO9XB/view?usp=drive_link

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1. Project Overview

This report presents the joint project prepared for the Computer Networks and Operating Systems courses. The project is a mobile learning application named LearNexus. It is designed for middle school students and aims to teach selected computing concepts through short explanations, everyday-life analogies, mini missions, quiz questions, feedback, profiles, badges, and progress tracking.

The application follows a guided learning flow. Instead of asking learners to answer questions immediately, it first introduces each subject through structured content. After the explanation part, students answer multiple-choice questions related to the same topic. This structure supports both conceptual understanding and active recall.

Project Identity

Item	Information
Application name	LearNexus
Team number	2
Target platform	Mobile devices
Development approach	React Native + Expo SDK 54
Target audience	Middle school students
Main language of application content	Turkish
GitHub repository	https://github.com/cemgedikoglu/LearNexus
Promotional video	https://drive.google.com/file/d/1mEOLP_ZhbSZsaX5AvIFu4BuZIVReO9XB/view?usp=drive_link

2. Course Topic Selection

The project required selecting one topic from the Computer Networks course schedule and one topic from the Operating Systems course schedule, then combining them in a single educational mobile application. The selected topics were chosen because they can be explained to middle school learners using concrete examples from daily digital life.

Course	Original Topic Name	Topic Used in the App	Course Schedule Relation	Reason for Selection
Computer Networks	Cybersecurity	Siber Güvenlik	Week 14 topic: Cybersecurity	Relevant to students' daily internet use, passwords, links, Wi-Fi, and safe behavior online.
Operating Systems	File-System Interface	Dosya Sistemi Arayüzü	Week 13 topic: File-System Interface	Easy to adapt with concrete examples such as files, folders, file extensions, and file paths.

The two selected topics were connected under the theme of digital literacy. Cybersecurity teaches students how to protect themselves online, while the file-system interface topic teaches them how digital information is organized and managed on a device.

3. Target Audience and Learning Objectives

The target group of the application is middle school students. Therefore, the content avoids highly technical terminology and presents each concept with familiar examples such as school bags, locks, folders, libraries, and addresses.

Computer Networks Learning Objectives

- Define cybersecurity in a simple and age-appropriate way.
- Recognize the features of a strong password.
- Explain the risk of phishing links and suspicious messages.
- Identify safer behaviors when using shared or public Wi-Fi networks.

Operating Systems Learning Objectives

- Explain what the file-system interface allows users to do.
- Distinguish between files and folders.
- Recognize common file extensions such as .jpg, .pdf, and .mp4.
- Understand that a file path shows the location of a file on a device.

4. Educational Design

The instructional model of the application is based on a simple sequence: learn, relate, apply, and check. Each lesson starts with explanation cards. These cards present the main concept, a short daily-life analogy, and a mini mission. After learners complete the explanation cards, they move to the quiz section.

Stage	Purpose	Example in the Application
Learn	Introduce the topic before questions.	Siber Güvenlik and Dosya Sistemi Arayüzü cards.
Relate	Connect abstract concepts to familiar situations.	Password as a lock; file path as an address.
Apply	Use a mini scenario to encourage decision-making.	Suspicious link scenario; organizing mixed files.
Check	Measure understanding with immediate quiz feedback.	Multiple-choice questions with correct/incorrect responses.

Why a Mobile Learning Design Was Used

The mobile learning design was selected because it supports short learning sessions, visible progress, and immediate feedback. The application first provides topic explanations, making it more suitable for middle school students who may be encountering these concepts for the first time.

5. Mobile Application Structure

The final application includes login, register, home, lesson, quiz, result, profile, badges, and side menu sections. The learner can enter the app, select a topic, read explanation cards, complete a mini mission, answer quiz questions, and view progress and badges.

Screen / Section	Function	Main Elements
Login Screen	Allows the user to enter the application.	Username, password, login button.
Register Screen	Allows the user to create a simple learning profile.	Name, username, password, register button.

Home Screen	Introduces the app and shows the learning missions.	App title, progress bar, lesson cards, quick actions.
Side Menu	Provides navigation inside the app.	Home, Profile, Badges, Exit.
Profile Screen	Shows learner information and progress.	Name, username, completed tasks, total score, badge count.
Badges Screen	Shows earned and locked badges.	Welcome badge, topic badges, completion badge.
Lesson Screen	Presents topic explanations before the quiz.	Concept card, analogy box, mini mission.
Quiz Screen	Checks student understanding.	Question, answer options, feedback message.
Result Screen	Shows performance after the quiz.	Score, completion message, progress result, badge navigation.

Learning Flow

Login/Register -> Home Screen -> Select Lesson -> Read Explanation Cards -> Complete Mini Mission -> Answer Quiz Questions -> View Result -> Check Profile and Badges -> Return Home or Exit

6. Technical Implementation

The project was implemented using React Native and Expo. This technology choice makes it possible to run the application on mobile devices through Expo Go without building a native Android or iOS package during the development stage. The final version was adjusted for Expo SDK 54 and tested on an iPhone using Expo Go.

Project File Structure

```
Team2_OS_ComNets/
|-- App.js
|-- app.json
|-- package.json
|-- package-lock.json
|-- README.md
|-- .gitignore
|-- index.js
|-- assets/
|  |-- logo.png
|-- data/
|   |-- lessons.js
```

Main Files

File	Role	Explanation
App.js	Main application file	Controls screen transitions, login/register flow, menu navigation, quiz logic, scoring, profile, badges, progress, and interface rendering.
data/lessons.js	Lesson data file	Stores lesson cards, objectives, missions, quiz questions, correct answers, and feedback texts.
package.json	Project configuration	Lists project scripts and dependencies required to run the

		Expo application.
app.json	Expo configuration	Defines the Expo application name, slug, version, orientation, and platform settings.
README.md	Project documentation	Explains the application, selected topics, setup process, promotional video, and final submission details.

Quiz and Progress Logic

- Each lesson contains a separate quiz array in the lesson data file.
- Each quiz question has options, a correct answer index, and a feedback message.
- When the learner selects an option, the app immediately checks whether the answer is correct.
- Correct answers increase the score, and the final result is displayed at the end of the lesson.
- Completed lessons are marked on the home screen to show overall progress.
- Badges are displayed based on login status and completed lessons.

7. User Interface and Interaction Design

The interface was designed to be simple, colorful, and appropriate for middle school students. Each lesson has its own color and icon, which helps students distinguish between the Computer Networks topic and the Operating Systems topic.

- Large titles and short paragraphs were used to reduce cognitive load.
- Progress bars were added to make learning progress visible.
- Cards and rounded buttons were used to create a mobile-friendly learning environment.
- Immediate feedback was included so that students can understand their mistakes quickly.
- The content is written in Turkish because the target learners are middle school students in Turkey.
- A side menu, profile screen, badges screen, and exit button were added according to project feedback.
- The application supports portrait and landscape orientation so that it can be used in different mobile screen positions.

8. Testing and Evaluation

The project was tested according to the main user flow of the application. The purpose of testing was to check whether the app can start correctly, display the login and register screens, show both lessons, move between screens, evaluate answers, display badges, and show the final result.

Test Case	Expected Result	Status
Start the app with <code>npx expo start -c</code>	Expo development server starts and QR code appears.	Completed
Open the app using Expo Go on iPhone	Login screen appears on the phone.	Completed
Use login and register screens	The user can enter the application.	Completed
Open the side menu	Menu items are displayed and navigation works.	Completed
Open profile and badges screens	User progress and badges are displayed.	Completed
Select Siber Güvenlik lesson	Lesson cards are displayed.	Completed
Select Dosya Sistemi Arayüzü lesson	Lesson cards are displayed.	Completed

Answer quiz questions	Feedback and score are shown.	Completed
Finish a lesson	Result screen and progress update are shown.	Completed
Rotate the device	The application supports portrait and landscape use.	Completed

The final promotional video was recorded on a real iPhone device and demonstrates the main features of the app. The video link is included in both the README file and this report.

9. Development Process and Group Collaboration

During the development process, all four group members participated in an online meeting through Google Meet. The project was implemented on a single computer to keep the development environment stable and avoid version conflicts.

Although the coding process was carried out on one computer, all group members contributed equally to the project. The group discussed the application structure, selected course topics, user interface decisions, learning flow, quiz content, feedback improvements, and final testing process together.

Based on the project feedback, the application was improved by adding login and register screens, a side menu, profile and badges sections, an exit button, portrait and landscape screen support, and fully Turkish topic names inside the application.

GitHub Repository Link: <https://github.com/cemgedikoglu/LearNexus>

Promotional Video Link: https://drive.google.com/file/d/1mEOLP_ZhbSZsaX5AvIFu4BuZIVReO9XB/view?usp=drive_link

10. Limitations and Future Work

The current version is a final educational prototype for course submission. It focuses on a clear learning flow, mobile usability, quiz interaction, profile tracking, and badges rather than advanced online account management or database storage.

Current Limitations

- The login and register screens are prototype-level and do not connect to a real database.
- Progress is stored only during the current app session.
- There is no teacher dashboard or online database.
- The prototype includes two lessons, one for each course topic.

Possible Future Improvements

- Add more lessons from both course schedules.
- Add animations or short interactive simulations for packet transfer and file organization.
- Store student progress locally or in a cloud database.
- Add more detailed badges, levels, and streaks to strengthen gamification.
- Add a teacher panel for viewing class progress.
- Add more interactive mini tasks and difficulty levels.

11. Conclusion

LearNexus combines two course topics into one mobile educational application. From Computer Networks, the project teaches Cybersecurity through the Turkish module Siber Güvenlik. From Operating Systems, it teaches File-System Interface through the Turkish module Dosya Sistemi Arayüzü. Both topics were adapted for middle school students using simple explanations, relatable examples, mini missions, and quizzes.

The project demonstrates how university-level computing topics can be simplified and transformed into an accessible learning experience for younger students. The final mobile structure, profile and badge features, side menu, orientation support, progress feedback, and quiz-based practice support an engaging and understandable learning process.

Appendix A. Running the Project

The project can be run on a computer with Node.js installed. After extracting the project folder, the following steps should be completed:

1. Open the Team2_OS_ComNets folder in Visual Studio Code.
2. Open the terminal inside the project folder.
3. Run `npm install` to install dependencies.
4. Run `npx expo start -c` to start the Expo development server with a clean cache.
5. Install Expo Go on a mobile phone.
6. Scan the QR code shown in the terminal.

Commands

```
npm install  
npx expo start -c
```

The project was tested with Expo SDK 54 and Expo Go on iPhone.

Appendix B. Repository, Video, and Submission Checklist

Before submission, the group should check the following requirements:

- The GitHub repository has been created and all final files have been pushed.
- All group members have joined the GitHub repository as collaborators.
- The GitHub repository link is included in this report.
- The promotional video link is included in this report.
- The promotional video link is accessible to anyone with the link.
- The report includes the names of all group members.
- The final ZIP file is renamed exactly as `Team2_OS_final.zip`.
- The final ZIP file size is under the upload limit or video is shared through a cloud link.
- Only one group submission is uploaded to `online.yildiz.edu.tr`.

GitHub Repository Link: <https://github.com/cemgedikoglu/LearNexus>

Promotional Video Link: https://drive.google.com/file/d/1mEOLP_ZhbSZsaX5AvIFu4BuZIVReO9XB/view?usp=drive_link